Vitor Gama

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Professional Summary

Chemical Engineer with extensive experience in process systems engineering, cost estimation, process simulation, optimization, and operability analysis. Skilled in coding and computational tools, bringing a strong technical edge to chemical engineering applications. A proactive and communicative team player, always open to discussing ideas, organizing events, and fostering a collaborative and engaging work environment. Strong leadership mindset with a focus on efficiency, problem-solving, and driving innovation through both technical expertise and team dynamics.

Technical Proficiencies

- **Programming:** Python, MATLAB, C#, JavaScript, LaTeX.
- **Process Simulation:** AVEVA Process Simulation, Aspen Plus, Aspen Custom modeler, HYSYS, PRO/II, ChemCad.
- **Version Control:** Git, GitHub, GitLab.
- **Languages:** English, Portuguese.

Education

West Virginia University — Morgantown, WV
Ph.D. candidate, Chemical Engineering — *Aug. 2021 – Dec. 2025 (Expected)*
GPA: 3.75/4.00

Federal University of Campina Grande — Campina Grande, Paraiba, Brazil
M.Sc., Chemical Engineering — *Sept. 2018 – April 2021*
GPA: 9.14/10.00

Federal University of Campina Grande — Campina Grande, Paraiba, Brazil
B.Sc., Chemical Engineering — *May 2013 – Aug. 2018*
GPA: 8.18/10.00

Awards & Achievements

- AIChE Environmental Division Graduate Paper Award (2024) – 2nd place.
- Jack and Marietta Mullenger Fellowship (2024)
- Statler College PhD Recruitment Fellowship (2021)

Research Experience

West Virginia University — Morgantown, WV
Graduate Research Assistant (Ph.D.) — *Aug. 2021 – Present*

- Design and simulation of gas separation membranes under Direct Air Capture (DAC) conditions.
- Applied inverse design and process operability analysis to optimize system performance.
- Published in peer-reviewed journals, demonstrating commitment to quality and research integrity.

Federal University of Campina Grande — Campina Grande, Paraiba, Brazil
Graduate Research Assistant (M.Sc.) — *Sept. 2018 – April 2021*

- Designed and implemented **CostApp**, a cost estimation tool, which enhances cost analysis and engineering decision-making efficiency.
- Developed process control routines in collaboration project with Petrobras, focusing on automation enhancements.

Research Publications

[1] **Vitor Gama** *et al.* *"Connecting Material Characteristics with System Properties for Membrane-based Direct Air Capture (m-DAC) using Process Operability and Inverse Design Approaches"*, **Industrial Engineering and Chemistry Research**. (April, 2025), DOI: 10.1021/acs.iecr.4c04553.

[2] **Vitor Gama** *et al.* *"Process Operability Analysis of Membrane-Based Direct Air Capture for Low-Purity CO_2 Production"*, **ACS Engineering Au**, 4.4 (Aug. 2024), pp. 394-404. DOI: 10.1021/acsengineeringau.3c00069.

[3] **Vitor V. Gama** *et al.* *"Modeling and Process Operability Analysis of a Direct Air Capture System"*, **IFAC-PapersOnLine**, 13th IFAC Symposium on Dynamics and Control of Process Systems, including Biosystems DYCOPS 2022, Vol. 55.7 (Jan. 2022), pp. 316-321. DOI: 10.1016/j.ifacol.2022.07.463. ISSN: 2405-8963.

Conference Presentations

[1] **Vitor Gama**, Oishi Sanyal, Fernando V. Lima. *Leveraging Process Operability Mapping to Support Experimental Membrane Direct Air Capture (m-DAC) Solutions.* 2024 AIChE Annual Meeting, San Diego, CA, October 29, 2024 (Oral Presentation)

[2] **Vitor Gama**, Oishi Sanyal, Fernando V. Lima. *Process Operability Analysis of Membrane-Based Direct Air Capture for Low-Purity CO_2 Production.* Membranes: Materials and Processes – Gordon Research Conference, New London, NH, July 28 – August 2, 2024 (Poster Presentation)

[3] **Vitor Gama**, Oishi Sanyal, Fernando V. Lima. *Evaluation of Hollow Fiber Membrane Modules for CO_2 Direct Air Capture and Utilization in Low-Purity CO_2 Processes.* 2023 AIChE Annual Meeting, Orlando, FL, November 7, 2023 (Oral Presentation)

[4] **Vitor Gama**, San Dinh, Oishi Sanyal, Fernando V. Lima. *Feasibility Analysis of a Membrane System for Direct Air Capture of CO_2.* 2022 AIChE Annual Meeting, Phoenix, AZ, November 17, 2022 (Oral Presentation)

[5] **Vitor Gama**, Oishi Sanyal, Fernando V. Lima. *Designing a Membrane-based Platform for CO_2 Direct Air Capture Systems Analysis.* Gordon Research Conference on Chemical Separations, Ventura, CA, October 2–7, 2022 (Poster Presentation)

Professional Leadership & Service

- President, Brazilian Student Association (2022 – 2024).
  - Helped students connect with other brazilians in town and engaged with the community during events by presenting elements of brazilian culture for the general population.
- Member, Statler College DEI Student Committee (2022 – 2023).
  - Promoted events that fostered a diversity on campus.
- Research Group Leader, CODES (2022 – Present).
  - Developed activities related to research group organization, lab visits and presentations.
- Research mentor, CODES (2022 – Present).
  - Mentored 3 undergraduate chemical engineering students, providing guidance in process simulation platforms, technical skills, and academic research development

CliftonStrengths

Arranger | Positivity | Input | Woo | Communication